All-phase Clifford atlas

24 projective classes / eight exact scalar phases / each distinct local alphabet

Each alphabet occupies eight pages. A page fixes k and lists all 24 operators $e^{ik\pi/4}C_j$. The representative matrices C_j are the exact named representatives in the catalog. Every displayed circuit equals its target literally; no scalar-phase repair or physical GP is included. The names identify actual catalog matrices, including the legacy distinction between SX and square-root X (Sx).
Shortest-path costs use the audited parameter-sensitive metric with nonnegative denominator penalties. A fixed Clifford alphabet such as H/S exhausts its allowed words. H/S/T/Tdg also has an exhaustive exact-word proof, including non-Clifford intermediate paths. For continuous alphabets the certificate covers the listed specializations and macro words. Missing paths retain that scope.
All presets with the same one-qubit alphabet share the same table. Entanglers and ancillary qubits are unavailable in this one-wire reference.

Exact representative matrices p. 2

H|S|T|Tdg p. 3

The 24 matrix representatives

These exact matrices fix the global-phase convention for every table.

C_0 / **I**

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

C_1 / **X**

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

C_2 / **Y**

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

C_3 / **Z**

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

C_4 / **H**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_5 / **S**

$$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

C_6 / **Sdg**

$$\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$

C_7 / **Sx**

$$\begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$$

C_8 / **Sxdg**

$$\begin{pmatrix} \frac{1-i}{2} & \frac{1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{pmatrix}$$

C_9 / **HX**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{10} / **HY_q2**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{11} / **HZ**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{12} / **SX_q3**

$$\begin{pmatrix} 0 & -\frac{\sqrt{2}(1+i)}{2} \\ \frac{\sqrt{2}(-1+i)}{2} & 0 \end{pmatrix}$$

C_{13} / **SY_q3**

$$\begin{pmatrix} 0 & \frac{\sqrt{2}(-1+i)}{2} \\ -\frac{\sqrt{2}(1+i)}{2} & 0 \end{pmatrix}$$

C_{14} / **HS**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{15} / **HSX_q6**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{16} / **HSY**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{17} / **HSdg**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{18} / **SH**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{19} / **SHX**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{20} / **SHY_q2**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{21} / **SHZ**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \end{pmatrix}$$

C_{22} / **SxY_q3**

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}i}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

C_{23} / **SxZ_q3**

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}i}{2} \\ -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

H|S|T|Tdg / $k = 0$

clifford_t_cx, clifford_t_cz

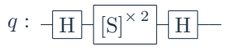
$e^{i0\pi/4}$ I

Score 0 / 0 gates



$e^{i0\pi/4}$ X

Score 8 / 4 gates



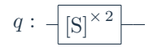
$e^{i0\pi/4}$ Y

Score 14 / 7 gates



$e^{i0\pi/4}$ Z

Score 4 / 2 gates



$e^{i0\pi/4}$ H

Score 2 / 1 gates



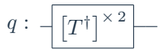
$e^{i0\pi/4}$ S

Score 2 / 1 gates



$e^{i0\pi/4}$ Sdg

Score 4 / 2 gates



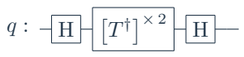
$e^{i0\pi/4}$ Sx

Score 6 / 3 gates



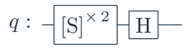
$e^{i0\pi/4}$ Sxdg

Score 8 / 4 gates



$e^{i0\pi/4}$ HX

Score 6 / 3 gates



$e^{i0\pi/4}$ HY_q2

Score 10 / 5 gates



$e^{i0\pi/4}$ HZ

Score 6 / 3 gates



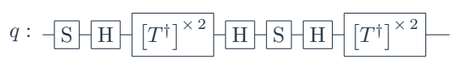
$e^{i0\pi/4}$ SX_q3

Score 18 / 9 gates



$e^{i0\pi/4}$ SY_q3

Score 18 / 9 gates



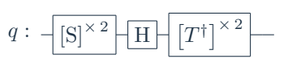
$e^{i0\pi/4}$ HS

Score 4 / 2 gates



$e^{i0\pi/4}$ HSX_q6

Score 10 / 5 gates



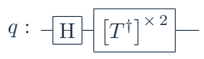
$e^{i0\pi/4}$ HSY

Score 8 / 4 gates



$e^{i0\pi/4}$ HSdg

Score 6 / 3 gates



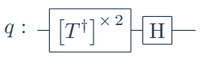
$e^{i0\pi/4}$ SH

Score 4 / 2 gates



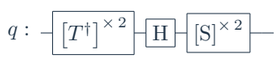
$e^{i0\pi/4}$ SHX

Score 6 / 3 gates



$e^{i0\pi/4}$ SHY_q2

Score 10 / 5 gates



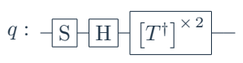
$e^{i0\pi/4}$ SHZ

Score 8 / 4 gates



$e^{i0\pi/4}$ SxY_q3

Score 8 / 4 gates



$e^{i0\pi/4}$ SxZ_q3

Score 20 / 10 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 1$

clifford_t_cx, clifford_t_cz

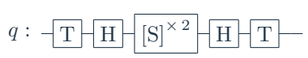
$e^{i1\pi/4}$ I

Score 12 / 6 gates



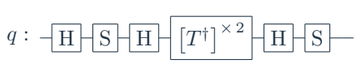
$e^{i1\pi/4}$ X

Score 12 / 6 gates



$e^{i1\pi/4}$ Y

Score 14 / 7 gates



$e^{i1\pi/4}$ Z

Score 14 / 7 gates



$e^{i1\pi/4}$ H

Score 10 / 5 gates



$e^{i1\pi/4}$ S

Score 14 / 7 gates



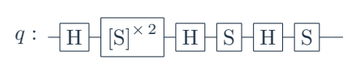
$e^{i1\pi/4}$ Sdg

Score 10 / 5 gates



$e^{i1\pi/4}$ Sx

Score 14 / 7 gates



$e^{i1\pi/4}$ Sxdg

Score 6 / 3 gates



$e^{i1\pi/4}$ HX

Score 12 / 6 gates



$e^{i1\pi/4}$ HY_q2

Score 14 / 7 gates



$e^{i1\pi/4}$ HZ

Score 12 / 6 gates



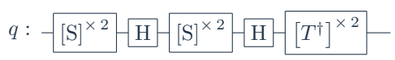
$e^{i1\pi/4}$ SX_q3

Score 16 / 8 gates



$e^{i1\pi/4}$ SY_q3

Score 16 / 8 gates



$e^{i1\pi/4}$ HS

Score 12 / 6 gates



$e^{i1\pi/4}$ HSX_q6

Score 10 / 5 gates



$e^{i1\pi/4}$ HSY

Score 14 / 7 gates



$e^{i1\pi/4}$ HSdg

Score 8 / 4 gates



$e^{i1\pi/4}$ SH

Score 12 / 6 gates



$e^{i1\pi/4}$ SHX

Score 8 / 4 gates



$e^{i1\pi/4}$ SHY_q2

Score 10 / 5 gates



$e^{i1\pi/4}$ SHZ

Score 14 / 7 gates



$e^{i1\pi/4}$ SxY_q3

Score 10 / 5 gates



$e^{i1\pi/4}$ SxZ_q3

Score 20 / 10 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

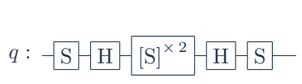
H|S|T|Tdg / $k = 2$

clifford_t_cx, clifford_t_cz

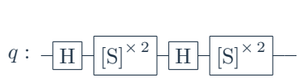
$e^{i2\pi/4}$ I
Score 20 / 10 gates



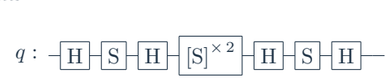
$e^{i2\pi/4}$ X
Score 12 / 6 gates



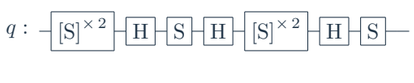
$e^{i2\pi/4}$ Y
Score 12 / 6 gates



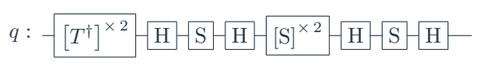
$e^{i2\pi/4}$ Z
Score 16 / 8 gates



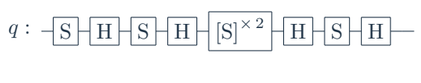
$e^{i2\pi/4}$ H
Score 18 / 9 gates



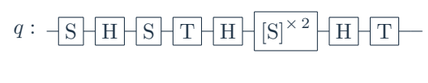
$e^{i2\pi/4}$ S
Score 20 / 10 gates



$e^{i2\pi/4}$ Sdg
Score 18 / 9 gates



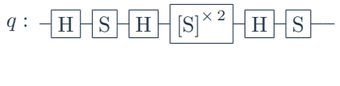
$e^{i2\pi/4}$ Sx
Score 18 / 9 gates



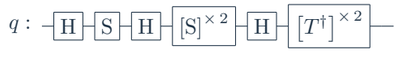
$e^{i2\pi/4}$ Sxdg
Score 14 / 7 gates



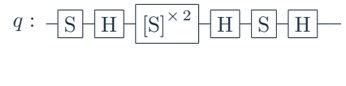
$e^{i2\pi/4}$ HX
Score 14 / 7 gates



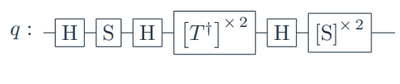
$e^{i2\pi/4}$ HY_q2
Score 16 / 8 gates



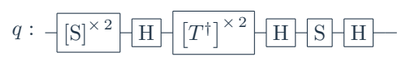
$e^{i2\pi/4}$ HZ
Score 14 / 7 gates



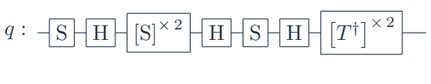
$e^{i2\pi/4}$ SX_q3
Score 16 / 8 gates



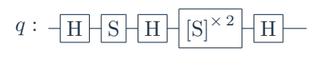
$e^{i2\pi/4}$ SY_q3
Score 16 / 8 gates



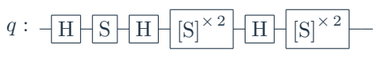
$e^{i2\pi/4}$ HS
Score 18 / 9 gates



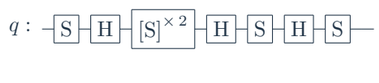
$e^{i2\pi/4}$ HSX_q6
Score 12 / 6 gates



$e^{i2\pi/4}$ HSY
Score 16 / 8 gates



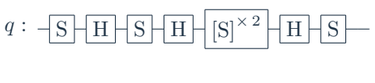
$e^{i2\pi/4}$ HSdg
Score 16 / 8 gates



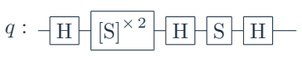
$e^{i2\pi/4}$ SH
Score 18 / 9 gates



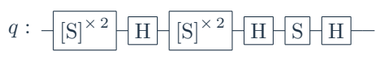
$e^{i2\pi/4}$ SHX
Score 16 / 8 gates



$e^{i2\pi/4}$ SHY_q2
Score 12 / 6 gates



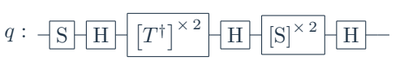
$e^{i2\pi/4}$ SHZ
Score 16 / 8 gates



$e^{i2\pi/4}$ SxY_q3
Score 16 / 8 gates



$e^{i2\pi/4}$ SxZ_q3
Score 16 / 8 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 3$

clifford_t_cx, clifford_t_cz

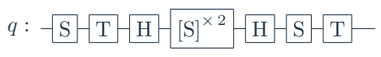
$e^{i3\pi/4}$ I

Score 24 / 12 gates



$e^{i3\pi/4}$ X

Score 16 / 8 gates



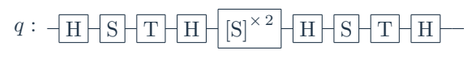
$e^{i3\pi/4}$ Y

Score 16 / 8 gates



$e^{i3\pi/4}$ Z

Score 20 / 10 gates



$e^{i3\pi/4}$ H

Score 22 / 11 gates



$e^{i3\pi/4}$ S

Score 22 / 11 gates



$e^{i3\pi/4}$ Sdg

Score 22 / 11 gates



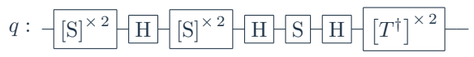
$e^{i3\pi/4}$ Sx

Score 18 / 9 gates



$e^{i3\pi/4}$ Sxdg

Score 20 / 10 gates



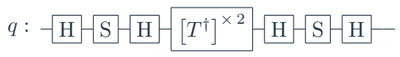
$e^{i3\pi/4}$ HX

Score 18 / 9 gates



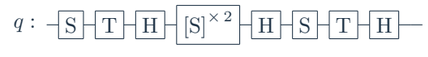
$e^{i3\pi/4}$ HY_q2

Score 16 / 8 gates



$e^{i3\pi/4}$ HZ

Score 18 / 9 gates



$e^{i3\pi/4}$ SX_q3

Score 12 / 6 gates



$e^{i3\pi/4}$ SY_q3

Score 12 / 6 gates



$e^{i3\pi/4}$ HS

Score 20 / 10 gates



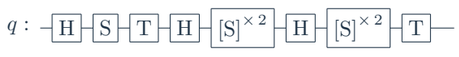
$e^{i3\pi/4}$ HSX_q6

Score 16 / 8 gates



$e^{i3\pi/4}$ HSY

Score 20 / 10 gates



$e^{i3\pi/4}$ HSdg

Score 20 / 10 gates



$e^{i3\pi/4}$ SH

Score 20 / 10 gates



$e^{i3\pi/4}$ SHX

Score 20 / 10 gates



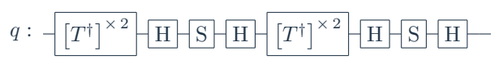
$e^{i3\pi/4}$ SHY_q2

Score 16 / 8 gates



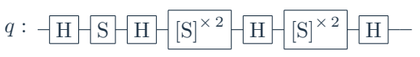
$e^{i3\pi/4}$ SHZ

Score 20 / 10 gates



$e^{i3\pi/4}$ SxY_q3

Score 18 / 9 gates



$e^{i3\pi/4}$ SxZ_q3

Score 12 / 6 gates



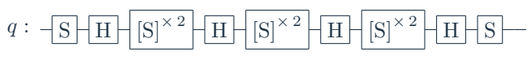
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 4$

clifford_t_cx, clifford_t_cz

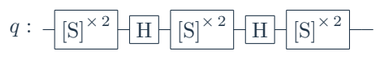
$e^{i4\pi/4}$ I

Score 24 / 12 gates



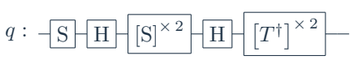
$e^{i4\pi/4}$ X

Score 16 / 8 gates



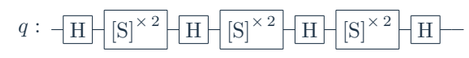
$e^{i4\pi/4}$ Y

Score 14 / 7 gates



$e^{i4\pi/4}$ Z

Score 20 / 10 gates



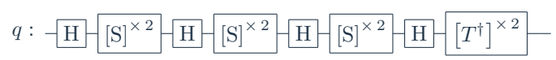
$e^{i4\pi/4}$ H

Score 22 / 11 gates



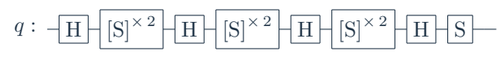
$e^{i4\pi/4}$ S

Score 24 / 12 gates



$e^{i4\pi/4}$ Sdg

Score 22 / 11 gates



$e^{i4\pi/4}$ Sx

Score 22 / 11 gates



$e^{i4\pi/4}$ Sxdg

Score 22 / 11 gates



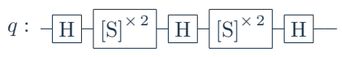
$e^{i4\pi/4}$ HX

Score 18 / 9 gates



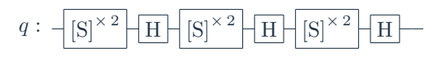
$e^{i4\pi/4}$ HY_q2

Score 14 / 7 gates



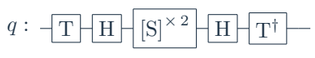
$e^{i4\pi/4}$ HZ

Score 18 / 9 gates



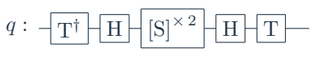
$e^{i4\pi/4}$ SX_q3

Score 12 / 6 gates



$e^{i4\pi/4}$ SY_q3

Score 12 / 6 gates



$e^{i4\pi/4}$ HS

Score 22 / 11 gates



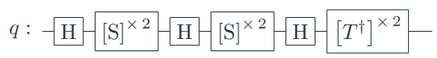
$e^{i4\pi/4}$ HSX_q6

Score 16 / 8 gates



$e^{i4\pi/4}$ HSY

Score 18 / 9 gates



$e^{i4\pi/4}$ HSdg

Score 20 / 10 gates



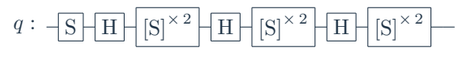
$e^{i4\pi/4}$ SH

Score 22 / 11 gates



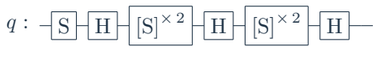
$e^{i4\pi/4}$ SHX

Score 20 / 10 gates



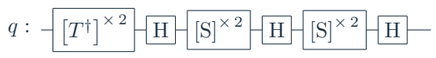
$e^{i4\pi/4}$ SHY_q2

Score 16 / 8 gates



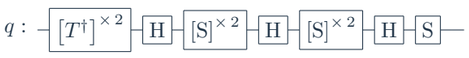
$e^{i4\pi/4}$ SHZ

Score 18 / 9 gates



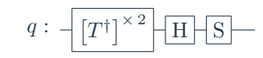
$e^{i4\pi/4}$ SxY_q3

Score 20 / 10 gates



$e^{i4\pi/4}$ SxZ_q3

Score 8 / 4 gates



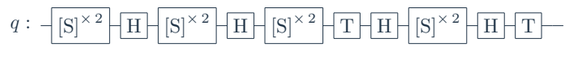
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 5$

clifford_t_cx, clifford_t_cz

$e^{i5\pi/4}$ I

Score 28 / 14 gates



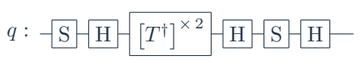
$e^{i5\pi/4}$ X

Score 20 / 10 gates



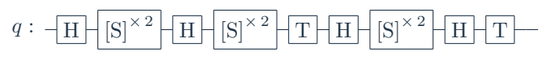
$e^{i5\pi/4}$ Y

Score 14 / 7 gates



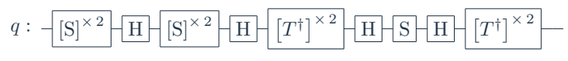
$e^{i5\pi/4}$ Z

Score 24 / 12 gates



$e^{i5\pi/4}$ H

Score 26 / 13 gates



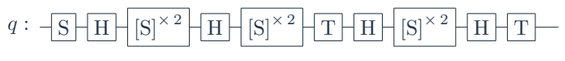
$e^{i5\pi/4}$ S

Score 24 / 12 gates



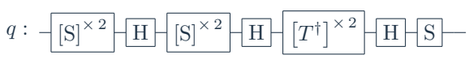
$e^{i5\pi/4}$ Sdg

Score 26 / 13 gates



$e^{i5\pi/4}$ Sx

Score 20 / 10 gates



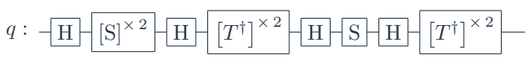
$e^{i5\pi/4}$ Sxdg

Score 22 / 11 gates



$e^{i5\pi/4}$ HX

Score 22 / 11 gates



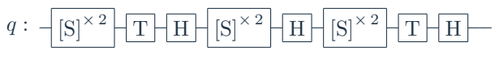
$e^{i5\pi/4}$ HY_q2

Score 18 / 9 gates



$e^{i5\pi/4}$ HZ

Score 22 / 11 gates



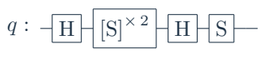
$e^{i5\pi/4}$ SX_q3

Score 10 / 5 gates



$e^{i5\pi/4}$ SY_q3

Score 10 / 5 gates



$e^{i5\pi/4}$ HS

Score 22 / 11 gates



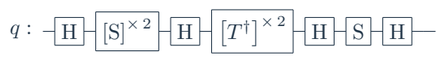
$e^{i5\pi/4}$ HSX_q6

Score 20 / 10 gates



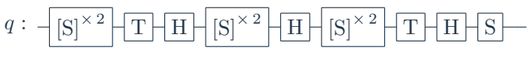
$e^{i5\pi/4}$ HSY

Score 18 / 9 gates



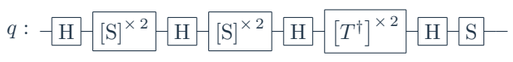
$e^{i5\pi/4}$ HSdg

Score 24 / 12 gates



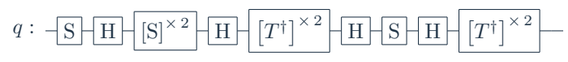
$e^{i5\pi/4}$ SH

Score 22 / 11 gates



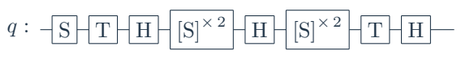
$e^{i5\pi/4}$ SHX

Score 24 / 12 gates



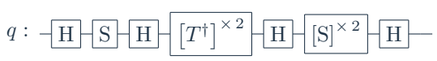
$e^{i5\pi/4}$ SHY_q2

Score 20 / 10 gates



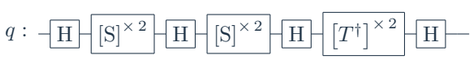
$e^{i5\pi/4}$ SHZ

Score 18 / 9 gates



$e^{i5\pi/4}$ SxY_q3

Score 20 / 10 gates



$e^{i5\pi/4}$ SxZ_q3

Score 10 / 5 gates



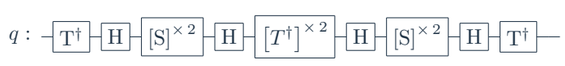
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 6$

clifford_t_cx, clifford_t_cz

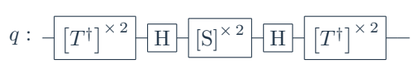
$e^{i6\pi/4}$ I

Score 24 / 12 gates



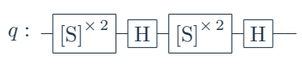
$e^{i6\pi/4}$ X

Score 16 / 8 gates



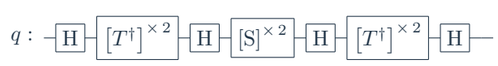
$e^{i6\pi/4}$ Y

Score 12 / 6 gates



$e^{i6\pi/4}$ Z

Score 20 / 10 gates



$e^{i6\pi/4}$ H

Score 22 / 11 gates



$e^{i6\pi/4}$ S

Score 20 / 10 gates



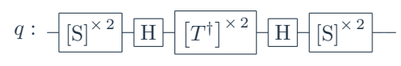
$e^{i6\pi/4}$ Sdg

Score 22 / 11 gates



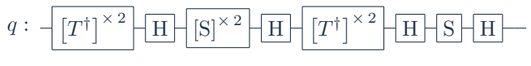
$e^{i6\pi/4}$ Sx

Score 16 / 8 gates



$e^{i6\pi/4}$ Sxdg

Score 22 / 11 gates



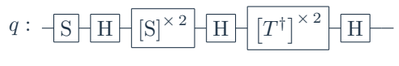
$e^{i6\pi/4}$ HX

Score 18 / 9 gates



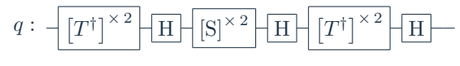
$e^{i6\pi/4}$ HY_q2

Score 16 / 8 gates



$e^{i6\pi/4}$ HZ

Score 18 / 9 gates



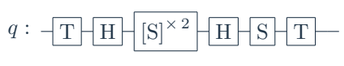
$e^{i6\pi/4}$ SX_q3

Score 14 / 7 gates



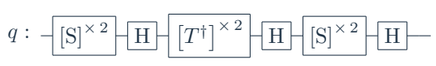
$e^{i6\pi/4}$ SY_q3

Score 14 / 7 gates



$e^{i6\pi/4}$ HS

Score 18 / 9 gates



$e^{i6\pi/4}$ HSX_q6

Score 18 / 9 gates



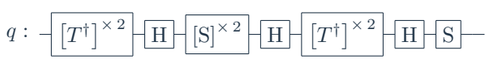
$e^{i6\pi/4}$ HSY

Score 14 / 7 gates



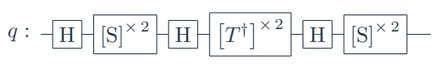
$e^{i6\pi/4}$ HSdg

Score 20 / 10 gates



$e^{i6\pi/4}$ SH

Score 18 / 9 gates



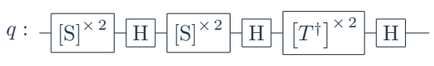
$e^{i6\pi/4}$ SHX

Score 20 / 10 gates



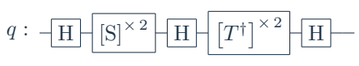
$e^{i6\pi/4}$ SHY_q2

Score 18 / 9 gates



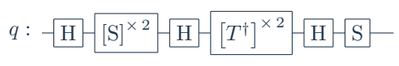
$e^{i6\pi/4}$ SHZ

Score 14 / 7 gates



$e^{i6\pi/4}$ SxY_q3

Score 16 / 8 gates



$e^{i6\pi/4}$ SxZ_q3

Score 16 / 8 gates



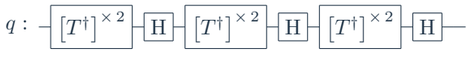
Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.

H|S|T|Tdg / $k = 7$

clifford_t_cx, clifford_t_cz

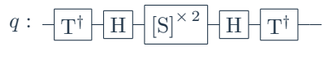
$e^{i7\pi/4}$ I

Score 18 / 9 gates



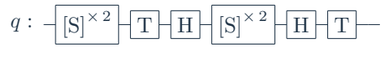
$e^{i7\pi/4}$ X

Score 12 / 6 gates



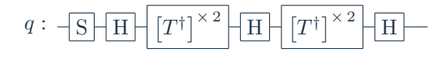
$e^{i7\pi/4}$ Y

Score 16 / 8 gates



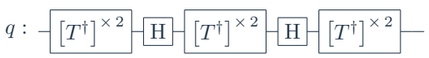
$e^{i7\pi/4}$ Z

Score 16 / 8 gates



$e^{i7\pi/4}$ H

Score 16 / 8 gates



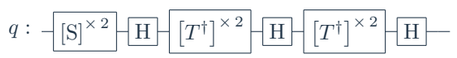
$e^{i7\pi/4}$ S

Score 14 / 7 gates



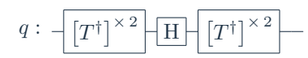
$e^{i7\pi/4}$ Sdg

Score 18 / 9 gates



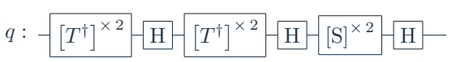
$e^{i7\pi/4}$ Sx

Score 10 / 5 gates



$e^{i7\pi/4}$ Sxdg

Score 18 / 9 gates



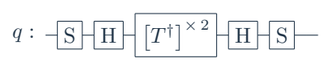
$e^{i7\pi/4}$ HX

Score 14 / 7 gates



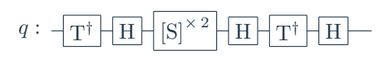
$e^{i7\pi/4}$ HY_q2

Score 12 / 6 gates



$e^{i7\pi/4}$ HZ

Score 14 / 7 gates



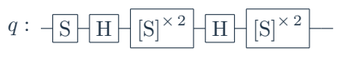
$e^{i7\pi/4}$ SX_q3

Score 14 / 7 gates



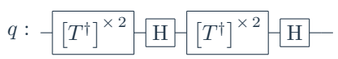
$e^{i7\pi/4}$ SY_q3

Score 14 / 7 gates



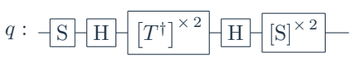
$e^{i7\pi/4}$ HS

Score 12 / 6 gates



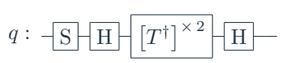
$e^{i7\pi/4}$ HSX_q6

Score 14 / 7 gates



$e^{i7\pi/4}$ HSY

Score 10 / 5 gates



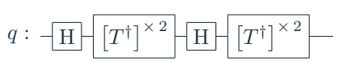
$e^{i7\pi/4}$ HSdg

Score 16 / 8 gates



$e^{i7\pi/4}$ SH

Score 12 / 6 gates



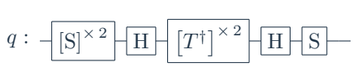
$e^{i7\pi/4}$ SHX

Score 16 / 8 gates



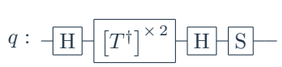
$e^{i7\pi/4}$ SHY_q2

Score 14 / 7 gates



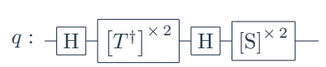
$e^{i7\pi/4}$ SHZ

Score 10 / 5 gates



$e^{i7\pi/4}$ SxY_q3

Score 12 / 6 gates



$e^{i7\pi/4}$ SxZ_q3

Score 18 / 9 gates



Certificate scope: all_words_in_fixed_H_S_T_Tdg_alphabet_including_nonclifford_intermediates. Phase is part of the target matrix.